

# Andrew Orfin

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## EDUCATION

### Stony Brook University

Stony Brook, NY

Bachelor of Science in Computer Science

May 2028

Double Major in Applied Mathematics and Statistics

GPA: 3.96

Relevant Coursework: *Analysis of Algorithms: Honors, Data Structures, System Fundamentals II, Principles of*

*Programming Languages, Fundamentals of Software Development, Linear Algebra, Survey of Probability and Statistics*

Honors/Scholarships: Computer Science Honors Program, University Scholars

## TECHNICAL SKILLS

**Languages:** JavaScript, TypeScript, C, Python, Java, SQL, SML, Prolog, HTML5, CSS3

**Frameworks & Libraries:** React, Node.js, Tailwind, ElizaOS, pandas, NumPy, Matplotlib, scikit-learn, Lex/Yacc, regex

**Tooling & Environments:** Git, Vite, Vitest, Docker, PostgreSQL (pgAdmin), Linux/POSIX, Bun, GDB, Valgrind, AddressSanitizer, VS Code, IntelliJ IDEA, Jupyter, Ollama, GitHub Copilot, OpenAI Codex, Arweave (Permaweb)

## EXPERIENCE

### DeScAI

Stony Brook, NY

Software Engineer

January 2026 - Present

- Engineered a back-end state-management workflow for *ElizaOS* AI framework utilizing *PostgreSQL*, *Docker*, and *SQL* migrations for automating polling, submission, & confirmation of *Arweave* transactions.
- Collaborated with a team of 5+ developers to debug, optimize, & deploy the DeScAI web front-end with *Vite*, *React*, & *TypeScript* while integrating secure blockchain-native comment publishing.
- Implemented data processing pipelines, mapping *Arweave* gateway responses into dynamic UI data structures for seamless customer-facing interface rendering from decentralized *Permaweb* storage.

## PROJECTS

### Semantic Analyzer, Independent Project

January 2026

- Architected a natural language processing pipeline in *Python* using *pandas* and *regex* to tokenize, cull, & process through TF-IDF vectorization 50k+ raw IMDb reviews.
- Fine-tuned a logistic regression training model with *scikit-learn* to predict review sentiment with 88%+ accuracy and *Matplotlib* to visualize model performance and analyze predictive token weights.

### POSIX/UNIX C Fuzzer, Class Project

April 2026

- Built a concurrent multi-process *C* fuzzer that automates software vulnerability testing for C files.
- Implemented a bidirectional inter-process communication framework through safe *POSIX* signal handling and shared memory verified safe by comprehensive unit tests designed with *Criterion*.
- Designed control-flow graph logic with bitmaps/hashsets for input discovery tracking across asynchronous runner processes and a mutation engine with 37 unique byte-level edge case alterations.

## LEADERSHIP

### Wolfie's Big Band

Stony Brook, NY

President

April 2025 - Present

- Directed operations, budgeting, and outreach for a Masquerade Ball of 200+ attendees, coordinating with 20+ student & local organizations to raise \$3,000+ for charitable causes.
- Orchestrated a 15+ member jazz band in weekly rehearsals, led logistics & performances for 10+ yearly live events averaging 100+ attendees.
- Devised executive meeting agendas, distributed directives, and trained a 5 person executive board.
- Cultivated an inclusive & supportive group culture through mentorship and monthly group bonding activities.